

Phase 2: Requirement Analysis

Technology Stack Justification

1. Technology Matrix Overview

Tier	Technology / Library	Version	Purpose in ComicCraft	Rationale & Justification
Backend Framework	FastAPI	>= 0.109.0	Server-side routing, request handling, form ingestion, and JSON REST API.	Exceptional async performance, automatic OpenAPI documentation (/docs), native Pydantic validation, and minimal boilerplate.
ASGI Server	Uvicorn	>= 0.27.0	High-performance asynchronous server running FastAPI.	Industry-standard lightning-fast ASGI server built on uvloop and httptools.
Template Engine	Jinja2	>= 3.1.2	Server-rendered dynamic HTML templates.	Seamless integration with FastAPI, expressive template syntax ({% for panel in layout %}), fast compilation.
LLM Orchestration	Google Gemini Flash (gemini-1.5-flash)	API / SDK	Generating structured 5-panel comic storyline outlines in JSON.	Extremely low latency, superior adherence to JSON schemas, cost-effective for rapid outlining.
Creative Storyteller	Google Gemini Pro (gemini-1.5-pro)	API / SDK	Synthesizing detailed comic narration, character dialogue, and captions.	Deep literary reasoning, nuanced character voice generation, and expressive dialogic pacing.
Visual Inking Engine	Stable Diffusion & Inking Pipeline	SD 1.5 / Diffusers	Generating comic-style illustrations from AI scene prompts.	High-quality visual fidelity, rich artistic styling support (anime, comic book, cyberpunk, realistic).
Document Compiler	FPDF2	>= 2.8.0	Compiling panels, illustrations, and text into a downloadable PDF document.	Lightweight pure-Python PDF generation with Unicode font embedding (DejaVuSans.ttf), precise millimeter layout control.
Image Processing	Pillow (PIL)	>= 10.0.0	Image format conversion, dimension checks, and procedural artistic synthesis.	Rock-solid Python imaging standard, fast raster processing, zero binary external dependencies.
Frontend Styling	HTML5 & Vanilla Modern CSS CSS3		Responsive comic book UI, glassmorphism, speech bubbles, and animations.	Maximum control, zero heavy UI frameworks, native CSS variables, instant loading speeds.
Configuration	python-dotenv	>= 1.0.0	Managing environment variables (GEMINI_API_KEY, HF_API_KEY).	Secure secret isolation preventing accidental credential leaks.

2. Comparative Analysis: Why FastAPI over Flask?

- Asynchronous Throughput: FastAPI's native async/await enables non-blocking handling of long-running generative AI calls.
- Schema Validation: FastAPI leverages Pydantic for strict runtime type enforcement (PromptRequest), preventing malformed requests.
- Interactive Documentation: Instant Swagger UI (/docs) and ReDoc (/redoc) for seamless developer and evaluator testing.